

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of M.Pharm (Pharmaceutical Technology)
University Theory Examination - April/May 2018
PHPT006- PHARMACEUTICAL PRODUCTION MANAGEMENT, TECHNOLOGY AND
DRUG REGULATORY AFFAIRS

Date: 30.04.18, Monday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** Write a note on Heating Ventilation and Air Conditioning System (HVAC) in context to pharmaceutical plant design. **5**
- B)** Describe in brief the hazards in pharmaceutical industries and its monitoring & prevention systems. **5**
- Q.2** **A)** What is IND application? Discuss the importance of Investigator's brochure in detail. **5**
- B)** How does MHRA regulate the marketing authorization of medicines? **5**

OR

- Q.2** **A)** State the purpose for the SUPAC guidelines. Describe in brief about SUPAC guidelines for change in manufacturing process of an immediate release dosage form. **5**
- B)** Write in brief on evaluation and stability aspects of packaging for liquid oral dosage forms. **5**
- Q.3** **Answer the following (Any three)** **15**
- (A)** Write a note on: Schedule Y of Drugs and Cosmetics Act, 1940 and rules.
- (B)** What are the requirements for product registration in India?
- (C)** What is ANDA? State the differences between NDA and ANDA.
- (D)** Explain the process scale up for semisolids.

SECTION-II

- Q.4** **A)** Discuss the application and role of robotics as an alternative to gowned personnel in context to advance Sterile Product Manufacturing Process and Technology. **5**
- B)** Describe the inspection procedure for inspection of pharmaceutical manufacturers according to WHO guidelines. **5**
- Q.5** Discuss ICH quality guidelines with respect to pharmaceutical development of a drug product. **10**

OR

- Q.5** Draw flowcharts along with In-process quality control tests showing manufacturing steps of large volume parenterals and dry powder injection. **10**
- Q.6** **Answer the following (Any three)** **15**
- (A)** Write a brief note on In-process quality control tests for hard gelatin capsule.
- (B)** Discuss importance of technology transfer and problems encountered during transfer of technology.
- (C)** Discuss role of market forecasting in production planning and inventory management.
- (D)** Write a short note on Industrial effluent testing and its treatment in context to control of environmental pollution.